

QD-MTS Research Platform

Quantum Dot-Assisted Diagnostic Prototype for Mesial Temporal Sclerosis

A computational research platform exploring the application of engineered quantum dot nanoparticles

to improve the accuracy and precision of MTS diagnosis, surgical planning, and fluorescence-guided treatment

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Prototype — Not for clinical diagnostic use

1. The Clinical Challenge: Mesial Temporal Sclerosis

Mesial temporal sclerosis (MTS) is the most common structural cause of drug-resistant focal epilepsy in adults. It is characterised by the selective loss of neurons in specific subregions of the hippocampus — primarily CA1, CA4, and the hilus — accompanied by reactive gliosis (scarring), mossy fibre sprouting into the dentate gyrus, and granule cell dispersion. These changes give rise to recurrent, unprovoked seizures originating in the medial temporal lobe.

Approximately 65 million people worldwide live with epilepsy; roughly one-third of these cases are drug-resistant, meaning standard antiepileptic medications fail to control seizures. MTS accounts for the majority of such refractory temporal lobe epilepsy cases. For those affected, uncontrolled seizures are not merely a medical inconvenience — they restrict independence, employment, education, and social participation, imposing a profound burden on daily life.

"Mesial temporal sclerosis affects millions of people globally. For many, it means years — sometimes decades — of uncontrolled seizures before a correct diagnosis is made and appropriate treatment is offered. The diagnostic gap is not a minor inefficiency. It is a period of measurable neurological harm."

1.1 The Diagnostic Problem

Despite being the single most common surgically treatable cause of epilepsy, MTS remains significantly underdiagnosed. The current standard of care relies on:

- High-resolution MRI (1.5T or 3T), which shows hippocampal atrophy and increased T2 signal.
- EEG with video monitoring, to localise the seizure onset zone.
- Neuropsychological testing and PET scanning in selected cases.

However, these tools have well-documented limitations. MRI misses MTS in 15–30% of cases — particularly in early-stage or subtle sclerosis where hippocampal volumetric loss is below the resolution threshold. The critical question of exactly which hippocampal subfields are affected, and to what severity, remains essentially invisible on clinical neuroimaging.

This matters enormously for treatment. Surgical outcomes (temporal lobectomy, radiofrequency ablation, or LITT — laser interstitial thermal therapy) depend critically on the precision of the target. Operating with incomplete diagnostic information increases the risk of incomplete resection of the epileptogenic zone, damage to eloquent cortex, and seizure recurrence post-surgery. Currently, between 30–40% of patients who undergo temporal lobe surgery for MTS continue to have seizures after the procedure.

Diagnostic Tool	Strength	Key Limitation
MRI (3T)	Non-invasive, widely available	Misses 15–30% of MTS; no subfield resolution
EEG / Video telemetry	Functional localisation of seizure onset	No structural or cellular information
PET / SPECT	Metabolic mapping	Low spatial resolution; expensive
Post-surgical histology	Definitive pathological confirmation	Only available after surgery; no pre-operative guidance

Table 1. Comparison of current diagnostic modalities for MTS with key limitations.

2. The Opportunity: Quantum Dot Nanoparticles

Quantum dots (QDs) are semiconductor nanocrystals, typically 2–10 nanometres in diameter, whose optical and electronic properties are exquisitely tunable by controlling their size and composition. Unlike conventional organic fluorescent dyes, quantum dots offer:

- Size-tunable emission wavelengths — the same material type emits different colours depending on particle size, enabling multi-channel imaging from a single synthesis platform.
- Near-infrared (NIR) emission capability — critical for brain tissue, which is most optically transparent in the 650–900 nm (NIR-I) and 1000–1700 nm (NIR-II) windows, allowing deeper tissue penetration with less background autofluorescence.
- High quantum yield and photostability — QDs resist photobleaching far better than organic dyes, enabling sustained, high-sensitivity imaging sessions.
- Versatile surface functionalisation — QDs can be conjugated to antibodies, peptides, aptamers, or small molecules, directing them to specific biological targets with molecular precision.
- Multi-modal potential — the same nanoparticle platform can, in principle, serve simultaneously as a diagnostic imaging agent, a drug carrier, and a biosensor.

These properties position quantum dots as uniquely suited to address the core diagnostic gap in MTS: the need to visualise specific cell populations — at molecular resolution — across the individual subfields of the hippocampus.

2.1 Why Quantum Dots for MTS Specifically?

MTS has a well-characterised cellular and molecular signature. The pathological changes follow predictable patterns that correspond directly to the known properties of targetable biomarkers:

MTS Pathology	Biomarker	Hippocampal Subfields	QD Targeting Strategy
Reactive astrogliosis	GFAP (upregulated)	CA1, CA3, CA4	Anti-GFAP antibody via EDC/NHS coupling
Neuronal loss	NeuN (depleted)	CA1, CA2, CA3, CA4	Anti-NeuN antibody, density mapping
Mossy fibre sprouting	Neuropeptide Y (NPY)	Dentate gyrus, CA3	Anti-NPY antibody conjugation
Granule cell dispersion	Prox1	Dentate gyrus	Anti-Prox1 antibody labelling
Neuroinflammation	Iba1 (microglia)	CA1, CA3, CA4	Anti-Iba1 dual-colour panel
Signalling dysregulation	mTOR / pS6	CA1, dentate gyrus	Anti-pS6 antibody (therapeutic relevance)

Table 2. MTS pathological features and corresponding quantum dot biomarker targeting strategy.

By engineering a panel of QDs — each conjugated to an antibody against one of these biomarkers and each emitting at a spectrally distinct NIR wavelength — it becomes possible to

simultaneously map the full pathological landscape of an MTS hippocampal section in a single imaging experiment. This would represent a fundamental advance over current histopathological methods, which rely on serial single-channel staining procedures.

3. The QD-MTS Research Platform

The QD-MTS Research Platform is a three-phase computational prototype designed to model, visualise, and optimise the application of quantum dot nanoparticles to the diagnosis of mesial temporal sclerosis. It is built as a self-contained interactive HTML application requiring no installation, running directly in any modern web browser.

The platform serves three interconnected purposes:

- To evaluate and rank candidate quantum dot materials against the specific diagnostic requirements of MTS — accounting for emission wavelength, biocompatibility, quantum yield, size, and ease of biological conjugation.
- To map the relationship between selected QD materials and the hippocampal biomarkers they would target, and to define a practical multi-colour labelling strategy for simultaneous detection of MTS pathology.
- To provide an interactive three-dimensional anatomical model of the hippocampus in which MTS severity patterns can be explored, compared against established ILAE classification criteria, and overlaid with simulated QD fluorescence to visualise where specific probes would accumulate.

Together, these three phases form a coherent argument: from material selection, through biological targeting, to visualised diagnostic outcome.

3.1 Phase 1 — Material Optimizer

The Material Optimizer presents seven candidate quantum dot systems alongside the properties most relevant to neurological tissue imaging. Users can adjust the relative importance of each diagnostic parameter through interactive sliders, and the platform recalculates a weighted score for each material in real time.

Candidate Materials

QD Material	Emission Range	Peak (nm)	QY (%)	Toxicity	Key Advantage
InP / ZnS	500–750 nm	~625	65	Low	Best Cd-free NIR-I candidate
CuInS ₂ / ZnS	550–950 nm	~720	50	Low	Broadest NIR-I coverage
Ag ₂ S	800–1200 nm	~1000	12	Low–Mod	NIR-II; deepest in vivo penetration
Silicon QDs	500–900 nm	~700	35	Very Low	Biodegradable; safest for in vivo
Carbon QDs	350–700 nm	~460	40	Very Low	Maximum biocompatibility
Graphene QDs	400–700 nm	~530	25	Very Low	Rich surface chemistry; biosensing

QD Material	Emission Range	Peak (nm)	QY (%)	Toxicity	Key Advantage
CdSe / ZnS	480–650 nm	~565	80	High (Cd ²⁺)	Benchmark QY; ex vivo reference only

Table 3. Quantum dot candidates with key optical and biological properties.

Scoring Parameters

Each material is scored across six parameters, each independently weighted by the user:

- Tissue Window (Emission): whether the emission peak falls within the NIR-I (650–900 nm) or NIR-II (1000–1700 nm) biological windows, which minimise brain tissue autofluorescence and maximise depth of signal detection.
- Quantum Yield: the efficiency of photon conversion — a direct measure of imaging brightness and detection sensitivity.
- Biocompatibility: the safety profile for neural tissue, weighted most heavily in in vivo mode due to the particular vulnerability of hippocampal neurons to heavy metal ion leaching.
- Size / BBB Penetration: the ability to cross the blood-brain barrier, which requires hydrodynamic diameter below approximately 10 nm and is essential for any in vivo diagnostic application.
- Photostability: resistance to photobleaching over sustained imaging periods.
- Conjugation Ease: the compatibility of the surface chemistry with standard antibody and peptide conjugation protocols (EDC/NHS, streptavidin-biotin, thiol-maleimide).

A toggle between Ex Vivo and In Vivo diagnostic modes automatically adjusts the default weight distribution to reflect the different constraints of each setting.

Key finding: In ex vivo mode (post-surgical biopsy imaging), InP/ZnS and CuInS₂/ZnS consistently rank highest. In in vivo mode, Silicon QDs and Ag₂S rise to the top due to their superior biocompatibility and BBB penetration. CdSe/ZnS, despite its unmatched quantum yield, ranks last in in vivo mode due to cadmium neurotoxicity.

3.2 Phase 2 — Targeting Strategy

Phase 2 translates the material selection from Phase 1 into a practical multi-colour labelling protocol. It defines a four-channel imaging panel in which each channel uses a spectrally distinct QD system conjugated to a different MTS biomarker, chosen to avoid spectral overlap and cover the full pathological range of the disease.

Ch.	QD System	Biomarker Target	Wavelength	Pathology Detected
1	InP / ZnS	NeuN (anti-NeuN)	~530 nm (green)	Neuronal density across CA1–CA4 — primary severity map
2	CuInS ₂ / ZnS	GFAP (anti-GFAP)	~720 nm (NIR)	Reactive astrogliosis — glial scarring extent
3	Silicon QDs	NPY + Prox1	~700 nm (far-red)	Mossy fibre sprouting + granule cell dispersion

Ch.	QD System	Biomarker Target	Wavelength	Pathology Detected
4*	Ag ₂ S	RVG-29 peptide	~1000 nm (NIR-II)	In vivo hippocampal targeting (future application)

Table 4. Four-channel QD labelling strategy for simultaneous MTS pathology mapping.

*Channel 4 is a future in vivo application; channels 1–3 are validated for ex vivo use.

The simultaneous use of three to four spectrally non-overlapping QD channels on a single hippocampal section would allow a pathologist or neurosurgeon to visualise neuronal loss, glial scarring, and aberrant axonal sprouting in a single imaging session — replacing what currently requires three separate staining protocols, each taking 24–48 hours.

Conjugation Protocols

Three conjugation strategies are relevant to the QD-MTS platform:

- EDC/NHS covalent coupling: Forms stable amide bonds between surface carboxyl groups on the QD and primary amines on antibody proteins. Suitable for InP/ZnS and CuInS₂/ZnS. Results in ~10–20% quantum yield reduction but maintains excellent antibody affinity.
- Streptavidin-biotin linkage: Non-covalent, ultra-high-affinity binding ($K_n \sim 10^{-15}$ M). Uses biotinylated primary antibody with streptavidin-coated QD. Slightly larger hydrodynamic size limits in vivo use but ideal for ex vivo multi-plex panels.
- Thiol-maleimide chemistry: Site-specific conjugation via cysteine residues on the targeting peptide. Used for RVG-29 peptide on PEG-coated Ag₂S and Silicon QDs. Preserves BBB-crossing capability for in vivo applications.

3.3 Phase 3 — 3D Hippocampal Model

Phase 3 provides an interactive three-dimensional anatomical model of the right hippocampus in coronal cross-section, rendered using Three.js WebGL technology. The model displays all six major hippocampal subfields in anatomically correct relative positions:

- CA1 (Sommer sector) — the longest arm of the cornu ammonis; most vulnerable to hypoxic injury and MTS-related neuronal loss.
- CA2 — a narrow transitional sector; relatively resistant to MTS-related degeneration.
- CA3 — the mossy fibre zone; moderately affected in classical MTS; contains the proximal axons of dentate granule cells.
- CA4 / Hilus — the hilar interneuron population; severely affected in ILAE Type 1a and Type 3 MTS.
- Dentate Gyrus — the site of mossy fibre sprouting and granule cell dispersion; the most epileptogenically active subfield in MTS.
- Subiculum — the transition zone between CA1 and the entorhinal cortex; relatively spared in most MTS subtypes.

Each subfield is rendered as a three-dimensional extruded mesh, colour-coded by severity on a continuous green (normal) to amber (moderate) to red (critical) scale. The model can be freely rotated in three dimensions by mouse drag, and zoomed by scroll.

ILAE Classification Presets

The ILAE (International League Against Epilepsy) has defined four subtypes of hippocampal sclerosis based on the predominant pattern of neuronal loss. The platform implements these as one-click presets:

- Type 1a (Classical MTS): Severe CA1 and CA4 neuronal loss with moderate CA3 involvement. The most common subtype (~60% of surgical cases). Associated with the best surgical outcomes.
- Type 1b (CA1-dominant): Severe CA1 loss with moderate CA4 involvement. Often associated with longer epilepsy duration before surgery.
- Type 2 (CA1 only): Isolated CA1 neuronal loss. The least severe pattern; associated with good post-operative seizure freedom rates.
- Type 3 (End folium): Hilar (CA4)-predominant sclerosis. Associated with a history of early febrile seizures.

In addition to the presets, individual subfield severity can be adjusted manually via sliders, allowing the visualisation of mixed or atypical MTS patterns that do not fit cleanly into the ILAE categories — common in real clinical practice.

QD Fluorescence Simulation

When a quantum dot system is selected in Phase 3, animated fluorescent particles appear on the hippocampal subfields that contain that probe's target biomarker. The particle density scales with subfield severity — a severely affected region with high GFAP expression will display a denser particle cloud than a mildly affected one.

This simulation provides an intuitive visualisation of what a QD-labelled hippocampal section would look like under NIR fluorescence microscopy, and how the signal distribution maps onto the three-dimensional anatomical structure of the sclerotic hippocampus.

4. Clinical Significance and Future Directions

The central argument of this platform is straightforward: the diagnostic gap in MTS — the inability to precisely characterise the extent and pattern of hippocampal sclerosis before surgery — is one of the main reasons why a significant minority of patients do not become seizure-free after temporal lobe surgery. Better diagnosis directly translates into better-targeted treatment and better outcomes.

A quantum dot-based multi-channel fluorescence panel, validated against post-surgical hippocampal biopsy tissue, could provide:

- Intraoperative guidance: Real-time fluorescence imaging of resection margins during temporal lobectomy, confirming that the sclerotic hippocampus has been completely removed.
- Pre-operative characterisation: More precise subfield-level MTS typing from stereotactic biopsy samples, improving surgical planning and patient selection.
- Research tool: A standardised, quantitative method for comparing MTS severity across patient cohorts in research studies, replacing subjective semi-quantitative histological scoring.
- Drug response monitoring: Serial ex vivo or (in future) in vivo QD imaging could track the effect of disease-modifying treatments on hippocampal gliosis and neuronal density over time.

4.1 Roadmap to Clinical Translation

Phase	Setting	Objective	Key QD Systems
Phase A (Near-term)	Ex vivo, fixed tissue	Validate 3-channel panel (NeuN, GFAP, NPY) on surgical biopsy sections from known MTS cases. Compare QD labelling against Timm silver stain and immunofluorescence gold standards.	InP/ZnS, CuInS ₂ /ZnS, Si QDs
Phase B (Medium-term)	Ex vivo, fresh frozen tissue; animal models	Optimise conjugation protocols. Develop intraoperative fluorescence probe for resection margin assessment. Test in rodent MTS models (kainic acid / pilocarpine).	CuInS ₂ /ZnS, Si QDs
Phase C (Long-term)	In vivo, pre-clinical	Validate BBB-crossing QD formulations (RVG-29 peptide + Si QDs or Ag ₂ S) in animal models. NIR-II in vivo hippocampal imaging. Safety and clearance studies.	Ag ₂ S, Si QDs

Phase	Setting	Objective	Key QD Systems
Phase D (Translational)	First-in-human, pre-surgical	Investigational use in MTS patients undergoing pre-surgical evaluation. Stereotactic biopsy + QD panel imaging to characterise sclerosis pattern prior to definitive surgery.	TBD (regulatory-approved formulation)

Table 5. Proposed roadmap for clinical translation of the QD-MTS diagnostic approach.

4.2 Challenges and Considerations

Translating this approach from computational prototype to clinical reality will require addressing several challenges:

- Regulatory pathway: Quantum dot formulations for in vivo human use will require extensive toxicology and biodistribution data before regulatory approval. Silicon QDs and carbon QDs, which metabolise to non-toxic products, are the most promising candidates for this pathway.
- Quantum yield in tissue: Biological matrices reduce effective quantum yield. Optimisation of surface coatings, ligand exchange, and conjugation protocols will be required to maintain sufficient signal-to-noise ratios in complex tissue environments.
- Spectral unmixing: Three-to-four-channel simultaneous QD imaging requires robust spectral unmixing algorithms to separate overlapping emission tails. This is a solved problem in the research setting but requires standardisation for clinical use.
- Detector infrastructure: NIR-II imaging (Ag₂S, ~1000 nm) requires InGaAs cameras not currently standard in clinical pathology or surgical suites. This limits the near-term translational path to NIR-I systems.
- Cost and access: Synthesis of high-quality, reproducible QD formulations at clinical scale requires significant investment in Good Manufacturing Practice (GMP) production infrastructure.

5. Conclusion

Mesial temporal sclerosis is a condition that has, for too long, been inadequately visible. The hippocampal damage that drives drug-resistant seizures is real, measurable, and — at the cellular level — well understood. What has been missing is a diagnostic technology capable of making that damage visible with the precision that modern neurosurgery demands.

Quantum dots, with their unique combination of tunable NIR emission, photostability, and targetable surface chemistry, offer a scientifically credible path toward that precision. The QD-MTS Research Platform presented here is the first step in articulating that path computationally: a tool for exploring, comparing, and visualising the material choices, biological targeting strategies, and anatomical outcomes that a QD-based MTS diagnostic system would involve.

This platform is not a clinical product. It is a thinking tool — a way of translating a scientific hypothesis into an interactive, explorable, shareable form. Its purpose is to communicate the feasibility and potential of the approach to researchers, clinicians, and funding bodies who may be in a position to take it further.

The path from this prototype to a validated clinical diagnostic tool is long and will require collaboration across materials science, neurology, neurosurgery, and regulatory science. But the scientific foundation is solid, the unmet clinical need is urgent, and the potential impact — for the millions of people living with drug-resistant MTS-related epilepsy — is profound.

Selected References

The following works informed the scientific basis of this prototype:

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This document is a companion to the interactive QD-MTS Research Platform simulation.

Not for clinical diagnostic use. Research prototype only.